

Results from Twin Studies of Other Disorders and Conditions

The twin study method has been used to try to determine the extent of genetic or environmental influence on a wide variety of traits and conditions. Among these are sense of humor, which appears to be largely environmentally determined, as MZ and DZ pairs have similar concordance. Examples of other diseases in which MZ concordance exceeds DZ concordance, suggesting a significant genetic component, include addictive behaviors such as cigarette smoking and alcohol drinking, mental illnesses such as schizophrenia, as well as stroke and certain types of high blood pressure. Twin studies of many other disorders are ongoing.

Conclusion

Twin studies provide a unique approach to investigating the determinants of a disease or condition. A single twin study cannot absolutely determine the importance of genetic or environmental factors. However, the twin study method, in combination with other approaches, can be a powerful tool for unraveling the causes of disease. SEE ALSO BEHAVIOR; FERTILIZATION; GENE AND ENVIRONMENT; GENE DISCOVERY; INHERITANCE PATTERNS.

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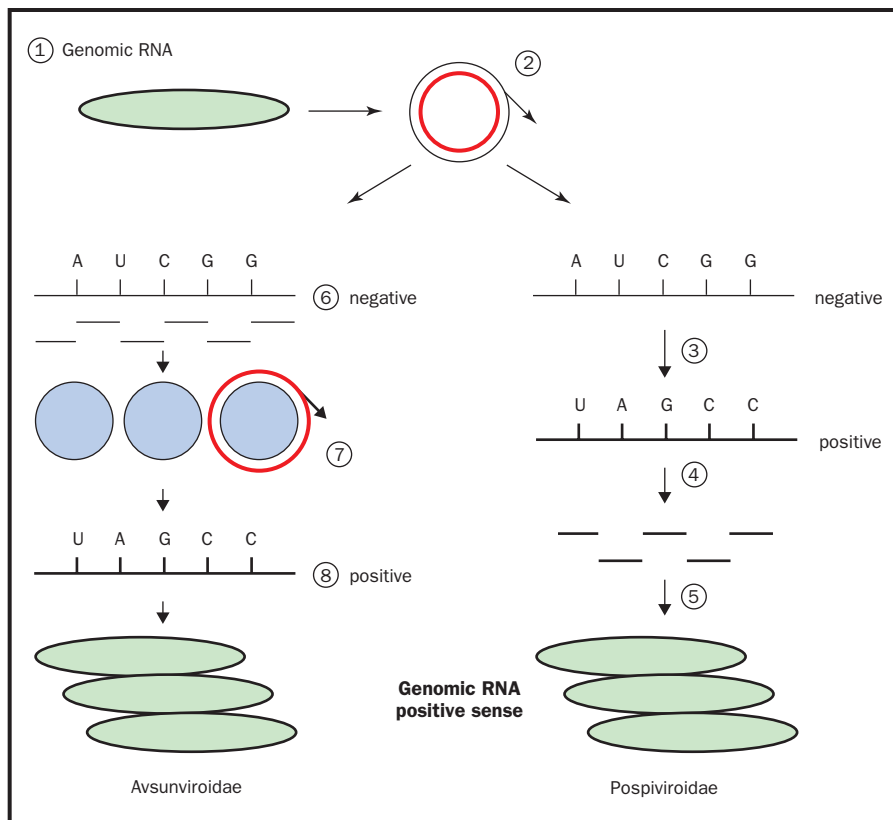
genome the total genetic material in a cell or organism

Viroids and Virusoids

Viruses are infectious agents consisting of a nucleic acid **genome** made of DNA or RNA, a protein coat, and sometimes lipids. They are able to replicate only inside cells, and the viral genome contains genes coding for proteins. Viroids and virusoids are also infectious agents, but they differ from viruses in several ways. For instance, they have a single-stranded circular, RNA genome. Their genomes are very small and do not code for proteins. Viroids replicate autonomously inside a cell, but virusoids cannot. Rather, virusoid replication requires that the cell is also infected with a virus that supplies "helper" functions.

Viroids

Viroids infect plant cells, and more than twenty-five kinds in two families are known. Viroid RNA is 246 to 375 nucleotides long and it folds to form rodlike structures with nucleotide base pairing (in which A pairs with U, C



Pospiviroidae, or viroid RNA (1), is copied by RNA polymerase (2), gives a negative strand first (3), and then a positive strand (4). This is cleaved and re-circularized to make viroid or virusoid RNA (5). Avsunviroidae, or virusoid RNA, is copied to make negative strand RNA (2, 6). Following ribozyme cleavage the negative circular RNA (7) is copied to make positive strand RNA (8). Ribozyme cleavage and recircularization results in formation of genomic RNA. Negative strand RNA is shown in thin lines, positive strand RNA is shown in thick lines.

pairs with G). The potato spindle tuber viroid and tomato plant macho viroid, both members of the family Pospiviroidae, replicate in the cell nucleus. A cellular enzyme, RNA polymerase, copies the circular RNA of the viroid to make a linear, repeated copy of the genome in **complementary** or “negative sense.” This RNA is copied again to make another linear, repeated “positive sense” RNA. Cellular enzymes cut this second copy of RNA at each place where the genome begins a repetition, yielding multiple copies of the genome. These copies then reassume a circular shape to make new viroid RNA.

The members of the second family of viroids, the Avsunviroidae, replicate in cell chloroplasts. Two are known, the avocado sunblotch viroid and the peach latent mosaic viroid. In both of these species, RNA polymerase makes a long, linear negative sense RNA. This RNA contains a catalytic **ribozyme** sequence, which cleaves itself. The negative sense RNA resumes its customary circular shape and is copied to form linear positive sense RNA. The ribozyme again cleaves this RNA, yielding linear genomic units that again recircularize, forming viroids.

The differing replication strategies of these two groups reflect different evolutionary origins. Viroids move through the plant in the phloem and plasmodesmata, which are part of the plant’s circulatory system. They propagate by mechanical means, vegetative reproduction, and possibly via seeds and insects. They cause plant diseases following interactions with proteins. For example, when they interact with an enzyme that impairs protein synthesis, the growth of the host plant may be stunted. This can have severe economic consequences.

complementary matching opposite, like hand and glove

ribozyme RNA-based catalyst

cytoplasm the material in a cell, excluding the nucleus

Virusoids

Virusoid genomes are 220 to 388 nucleotides long. A virusoid genome does not code for any proteins, but instead serves only to replicate itself. Virusoids can replicate in the **cytoplasm** and possess a ribozyme activity. RNA replication is similar to that of viroids, but each requires that the cell be infected with a specific “helper” virus. Five virusoids are known, and the helper viruses for these are all members of the Sobemovirus family. An example of a “helper” virus is the subterranean clover mottle virus, which has an associated virusoid. Virus enzymes may aid replication of the virusoid RNA. The virusoid is incorporated into the virus particle and transmitted as a “satellite,” a separate nucleic acid not part of the viral chromosome. Replication of the helper virus is independent of the virusoid.

Virusoids belong to a larger group of infectious agents called satellite RNAs, which are found in bacteria, plants, fungi, invertebrates, and vertebrates. Some satellite RNAs encode proteins; these are called satellite viruses and, like virusoids, must coinfect with a helper virus in order to replicate. One satellite RNA infecting humans is the hepatitis delta virusoid. It has a circular, single-stranded RNA genome of 1,700 nucleotides. Its helper is the hepatitis B virus, which is associated with liver disease. Coinfection with both agents results in a more severe infection. SEE ALSO RIBOZYME; RNA POLYMERASES; VIRUS.

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Virus

virion virus particle

A virus is a parasite that must infect a living cell to reproduce. Although viruses share several features with living organisms, such as the presence of genetic material (DNA or RNA), they are not considered to be alive. Unlike cells, which contain all the structures needed for growth and reproduction, viruses are composed of only an outer coat (capsid), the genome, and, in some cases, a few enzymes. Together these make up the **virion**, or virus particle. Many illnesses in humans, including AIDS, influenza, Ebola fever, the common cold, and certain cancers, are caused by viruses. Viruses also exist that infect animals, plants, bacteria, and fungi.

Physical Description and Classification

Viruses are distinguished from free-living microbes, such as bacteria and fungi, by their small size and relatively simple structures. Diminutive viruses such as parvovirus may have a diameter of only 25 nanometers (nm, 10^{-9} meters). Poxviruses, the largest known viruses, are about 300 nanometers across, just at the detection limits of the light microscope. Typical bacteria have diameters of 1,000 nanometers or more. Information on the structure of viruses has been obtained with several techniques, including electron

Nucleic Acid	Polarity	Family	Examples	Host	Diseases/pathologies
ss DNA	+	<i>Parvoviridae</i>	parvovirus B19	humans	erythema infectiosum (fifth disease)
ds DNA	+/-	<i>Myoviridae</i>	Bacteriophage T4	<i>E. coli</i>	bacterial lysis
		<i>Papillomaviridae</i>	HPV types 2, 16, 18, 33	humans	warts, cervical and other cancers
		<i>Herpesviridae</i>	herpes zoster virus	humans	chicken pox, shingles
		<i>Poxviridae</i>	variola virus	humans	smallpox
ss RNA non-seg.	+	<i>Picornaviridae</i>	poliovirus types 1-3	humans	poliomyelitis
			rhinovirus (100+ serotypes)	humans	common cold
		<i>Togaviridae</i>	equine encephalitis virus	insects/horses	CNS disease in horse and humans
ss RNA non-seg.	-	<i>Rhabdoviridae</i>	rabies virus	mammals	rabies
		<i>Paramyxoviridae</i>	measles virus	humans	measles
ssRNA segmented	-	<i>Orthomyxoviruses</i>	influenza virus	mammals, birds	influenza
ssRNA segmented	- and/or ambi	<i>Bunyaviridae</i>	Sin Nombre virus	rodents	hanta fever
		<i>Arenaviridae</i>	Lassa fever virus	primates	hemorrhagic fever
ds RNA	+/-	<i>Reoviridae</i>	Rice dwarf virus	plants	stunting
ssRNA DNA rep. int.	+	<i>Retroviridae</i>	HIV types 1, 2	humans	AIDS
			HTLV type I	humans	adult T-cell leukemia
ds DNA +/- RNA rep. int.	+/-	<i>Hepadnaviridae</i>	hepatitis B virus	humans	hepatitis, hepatocellular carcinoma

ss=single-stranded; ds=double stranded; non-seg.= non-segmented; ambi = ambisense; rep. int = replicative intermediate; HPV= human papillomavirus; CNS = central nervous system.

microscopy (EM). The limit of resolution of traditional EM is about 5 nm. With advanced EM techniques, such as cryogenic EM (cryoEM, in which the sample is rapidly frozen instead of exposed to chemical fixatives), coupled with computer image processing, smaller structures (1–2 nm) can be resolved. However, X-ray crystallography is the only method that allows for atomic-level resolution. Small viruses that produce uniform particles can be crystallized. The first atomic-level structure of a virus, tomato bushy stunt virus, was solved in 1978.

There is great diversity among viruses, but a limited number of basic designs. Capsids are structures that contain the viral genomes; many have icosahedral symmetry. An icosahedron is a three-dimensional, closed shape composed of twenty equilateral triangles. Viral proteins, in complexes termed “capsomers,” form the surface of the icosahedron.

Other viruses, such as the virus that causes rabies, are helical (rod shaped). The length of helical viruses can depend on the length of the genome, the DNA or RNA within, since there are often regular structural interactions between the nucleic acids of the genome and the proteins that cover it.

A **lipid**-containing envelope is a common feature of animal viruses, but uncommon in plant viruses. Embedded in the envelope are surface proteins, usually **glycoproteins** that help the virus interact with the surface of the cell it is infecting. A matrix layer of proteins often forms a bridge between the surface glycoproteins and the capsid. Some viruses, such as the picornaviruses, are not enveloped, nor do they have a matrix layer. In these viruses, cell-surface interactions are mediated by the capsid proteins.

Some viruses have compound structures. The head of the T4 bacterial virus (bacteriophage) is icosahedral and is attached via a collar to a contractile

Table 1. Classification of selected viruses by nucleic acid replication strategy (Baltimore Scheme).

lipid fat or waxlike molecule, insoluble in water

glycoproteins proteins to which sugars are attached

Table 2. The polarity, or sense, of a strand is an indication of how its sequence relates to messenger RNA (mRNA).

Molecule	Sequence	Polarity or Sense
Complementary RNA	A U U G G G C U C	negative
Coding strand DNA	T A A C C C G A G	positive
Complementary DNA	A T T G G G C T C	negative
mRNA	U A A C C C G A G	positive

antigenicity ability to provoke an immune response

reverse transcriptase enzyme that copies RNA into DNA

tail with helical symmetry. Large viruses, such as the herpesviruses and poxviruses, can have higher-ordered and more complex structures.

Classification of viruses considers the genome characteristics, virion shape and macromolecular composition, and other properties, such as **antigenicity** and host range. A scheme for classification of viruses based on the type of nucleic acid (DNA or RNA) present in the virus particle and the method of genome replication was devised by David Baltimore, co-discoverer of **reverse transcriptase** (see Table 1). Reverse transcriptase is an enzyme that converts retroviral genomic single-stranded (ss) RNA into doubled-stranded (ds) DNA.

Viral genomes can be RNA or DNA, positive or negative in polarity, ss or ds, and one continuous (sometimes circular) molecule or divided into segments. By convention, messenger RNA (mRNA) that can be directly translated to protein is considered positive sense (or positive in polarity). DNA with a corresponding sequence (that is, the coding strand of double-stranded DNA) is also a positive-sense strand. An RNA or DNA molecule with the reverse complementary sequence to mRNA is a negative-sense strand. A few viruses have been identified that contain one or more “ambisense” genomic RNA segments that are positive sense in one part of the molecule (this part can be translated directly into protein) and negative sense (reverse complement of coding sequence) in the rest of the molecule.

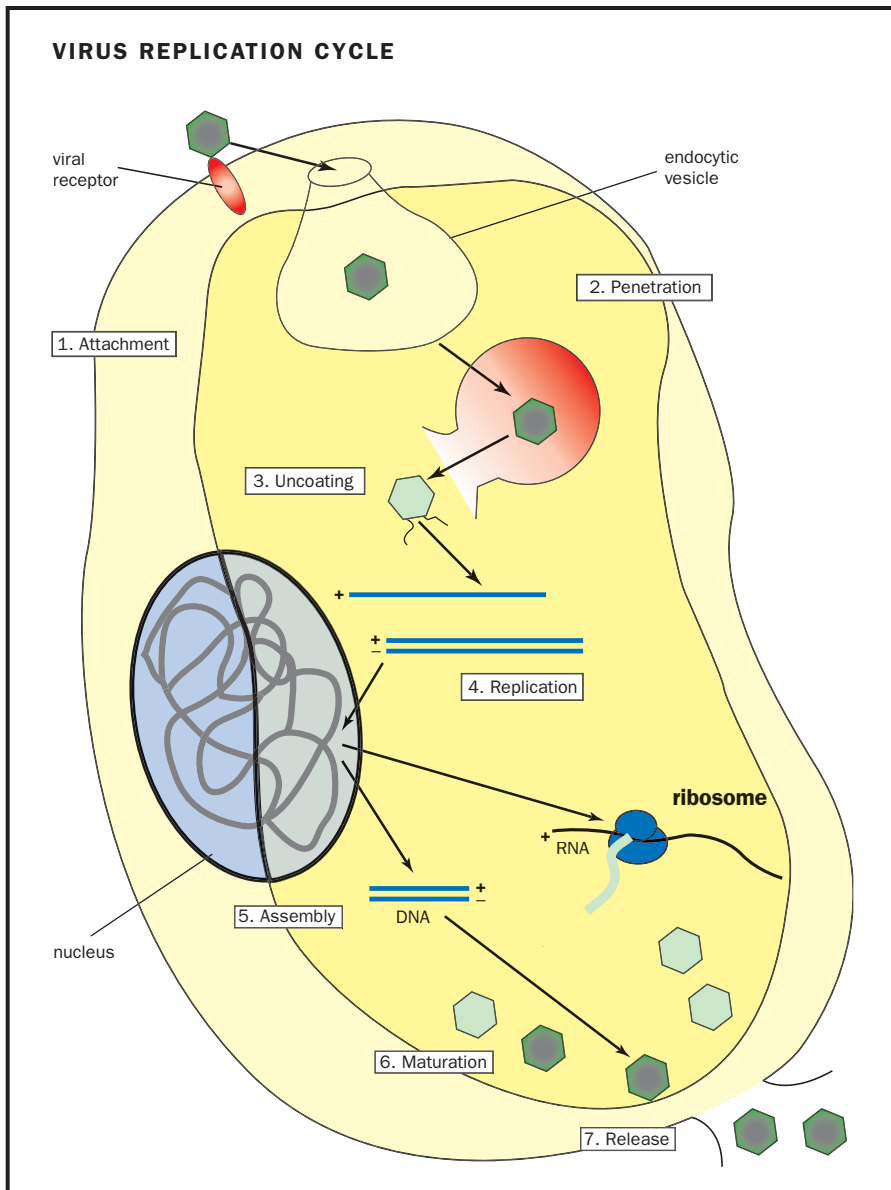
Virus Replication Cycle

For a virus to multiply it must infect a living cell. All viruses employ a common set of steps in their replication cycle. These steps are: attachment, penetration, uncoating, replication, assembly, maturation, and release.

Attachment and Penetration. A virion surface protein must bind to one or more components of the cell surface, the viral receptors. The presence or absence of receptors generally determines the type of cell in which a virus is able to replicate. This is called viral tropism. For example, the poliovirus receptor is present only on cells of higher primates and then in a limited subset of these, such as intestine and brain cells. While called virus receptors, these are actually used by the cell for its own purposes, but are exploited by the virus for entry.

Entry of the viral genome into the cell can occur by direct penetration of the virion at the cell surface or by a process called endocytosis, which is the engulfment of the particle into a membrane-based vesicle. If the latter, the virus is released when the vesicle is acidified inside the cell. Enveloped viruses may also fuse with the cellular surface membrane, which results in release of the capsid into the **cytoplasm**. Surface proteins of several viruses

cytoplasm the material in a cell, excluding the nucleus



Virus attaches to a receptor, enters the cell, and is uncoated, releasing its genetic material. In this example, positive polarity (+) DNA is released, which triggers formation of a complementary (–) DNA strand. Double-stranded DNA is copied in the nucleus to make progeny DNA. It is also transcribed to make messenger RNA, for synthesis of viral proteins at the ribosome.

contain “fusion peptides,” which are capable of interacting with the lipid bilayers of the host cell.

Uncoating and Replication. After penetration, viral capsid proteins must be removed, at least partly, to express and replicate the viral genome. In the case of most DNA viruses, the capsid is routed to the nucleus prior to uncoating. An example can be seen in the poxviruses, whose large DNA genomes encode most of the proteins needed for DNA replication. These viruses uncoat and replicate completely in the cytoplasm. RNA viruses typically lose the protective envelope and capsid proteins upon penetration into the cytoplasm. In reoviruses, only an outer protein shell is removed and replication takes place inside a structured subviral particle.

Viral genomes must be expressed as mRNAs in order to be translated into structural proteins for the capsids and, in some cases, as replicative proteins for replicating the virus genome. Viral genomes must also provide templates

that can be replicated to produce progeny genomes that will be packaged into newly produced virions. Replication details vary among the different types of viruses.

The ss positive-sense DNA of parvoviruses is copied by host DNA polymerase (the enzyme that replicates DNA) in the nucleus into a negative-sense DNA strand. This in turn serves as a template for mRNA and progeny DNA synthesis. The genomes of larger DNA viruses, with the exception of the poxviruses, are also transcribed and replicated in the nucleus by a combination of viral and host enzymes (for example, DNA-dependent RNA polymerases for transcription of mRNAs, DNA-dependent DNA polymerase for genome replication).

Positive-polarity RNA virus genomes can be translated directly, but for effective progeny production additional rounds of RNA replication via a negative-stranded intermediate are required. This is accomplished by a viral transcriptase (RNA-dependent RNA polymerase) and associated cofactors. Single-stranded negative-sense RNA viruses of animals must also carry a viral transcriptase to transcribe functional mRNAs and subsequently produce proteins, since this RNA-to-RNA enzymatic activity is typically lacking in animal cells.

Retroviruses are unique among viruses in that the genome is diploid, meaning that two copies of the positive-polarity RNA genome are in each virus particle. The genomic RNA is not translated into protein, but rather serves as a template for reverse transcription, which produces a double-stranded DNA via a viral reverse transcription enzyme. The DNA is subsequently integrated into the host cell chromosomal DNA. Hepadnaviruses also encode a reverse transcriptase, but replication occurs inside the virus particle producing the particle-associated genomic DNA.

Assembly, Maturation, and Release. As viral proteins and nucleic acids accumulate in the cell, they begin a process of self-assembly. Viral self-assembly was first demonstrated in a seminal series of experiments in 1955, wherein infectious particles of tobacco mosaic virus spontaneously formed when purified coat protein and genomic RNA were mixed. Likewise, poliovirus capsomers are known to self-assemble to form a procapsid in the cytoplasm. Progeny positive-strand poliovirus RNAs then enter this nascent particle. “Chaperone” proteins (chaperonins) of the cell play a critical role in facilitating the assembly of some viruses. Their normal role is to help fold cellular proteins after synthesis.

The maturation and release stages of the replication cycle may occur simultaneously with the previous step, or may follow in either order. Many viruses assemble their various components into “immature” particles. Further intracellular or extracellular processing is required to produce a mature infectious particle. This may involve cleavage of precursors to the structural proteins, as in the case of retroviruses.

Viruses that are not enveloped usually depend upon disintegration or **lysis** of the cell for release. Enveloped viruses can be released from the cell by the process of budding. In this process the viral capsid and usually a matrix layer are directed to a modified patch of cellular membrane. Interactions between the matrix proteins and/or envelope proteins drive envelopment. In the case of viruses that bud at the cell surface, such as some

lysis breakage

togaviruses and retroviruses (including HIV), this also results in release of the virus particles. If the virus acquires a patch of the nuclear membrane (as is the case with herpesviruses), then additional steps involving vesicular transport may be required for the virus to exit the cell.

Infection Outcomes

Viral infection can result in several different outcomes for the virus and the cell. Productive infection, such that each of the seven steps outlined above occurs, results in the formation of progeny viruses. Cells productively infected with poliovirus can yield up to 100,000 progeny virions per cell, although only a small fraction (fewer than 1 per 1,000) of these are capable of going on to carry out a complete replication cycle of their own. Productive infection may induce cell lysis, which results in the death of the cell. Nonenveloped viruses typically induce cell lysis to permit release of progeny virions. Many enveloped viruses also initiate events that result in cell death by various means, including **apoptosis**, **necrosis**, or lysis.

Viral infection may be abortive, in which one or more necessary factors, either viral or cellular, are absent and progeny virions are not made. Infection may be nonproductive, at least transiently, but viral genomes may still become resident in the host cell. Herpesviruses and retroviruses can establish **latent** infections. Latently infected cells may express a limited number of viral products, including those that result in cell transformation. Latent infections can often be activated by various stimuli, such as stress in the case of herpesviruses, to undergo a productive infection.

Viral Cancers

Infection with certain viruses can also result in cell transformation, stable genetic changes in the cell that result in disregulated cell growth and extended growth potential (immortalization). In animals, such virally induced cellular changes can result in cancer. This correlation was first made by Harry Rubin and Howard Temin in the 1950s, when they observed that Rous sarcoma virus, a retrovirus capable of inducing solid tumors in chickens, could also cause biochemical and structural changes and extend the proliferative potential of cultured chicken cells.

Viruses are perhaps second only to tobacco as risk factors for human cancers. DNA tumor viruses include papillomaviruses and various herpes viruses (such as HHV-8, which causes Kaposi's **sarcoma**). More than sixty strains of human papillomaviruses (HPV) have been identified. HPV cause warts, which are benign tumors, but are also the causes of **malignant** penile, vulval, and cervical cancers. Infection with hepatitis B or C viruses is associated with increased incidence of liver cancer. Adenoviruses have been shown to induce cancers in animals, but not in humans. Retroviruses can also cause cancer in various animal species, including humans. HTLV-1 causes adult T-cell leukemia in about 1 percent of infected humans.

Viruses can cause cancer through their effects on two important cellular genes or gene products: tumor suppressors and **oncogenes**. These genes are critical players in cell-cycle regulation. One protein product from HPV binds to the retinoblastoma (Rb) tumor suppressor protein. HPV E6 protein binds p53 tumor suppressor protein and promotes its degradation.

apoptosis programmed cell death

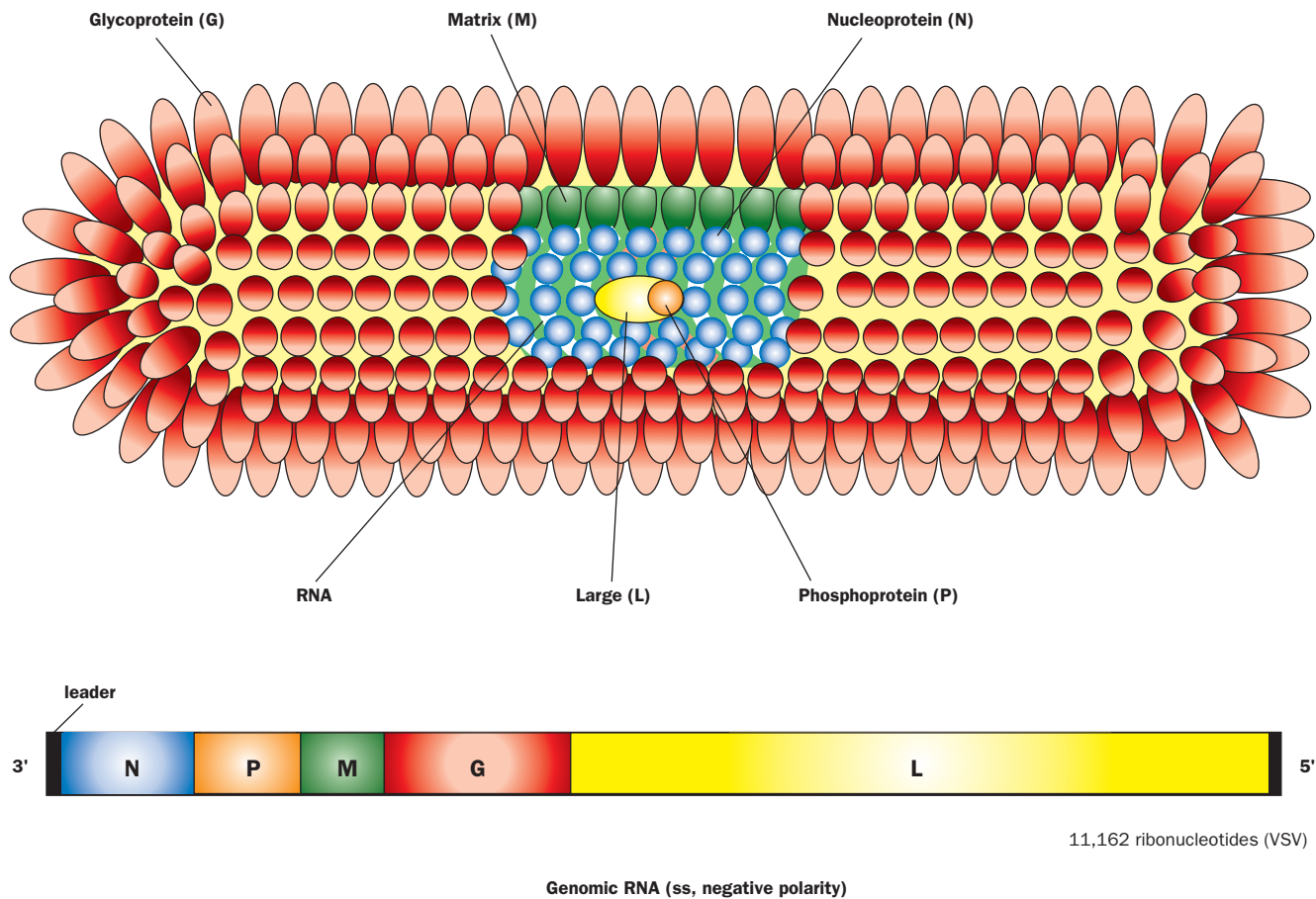
necrosis cell death from injury or disease

latent present or potential, but not apparent

sarcoma a type of malignant (cancerous) tumor

malignant cancerous; invasive tumor

oncogenes genes that cause cancer

RHABDOVIRUS (RABIES VIRUS, VSV)

Rhabdoviruses are helical viruses and include the virus that causes rabies and vesicular stomatitis virus (VSV), a common laboratory virus. The lipid-containing envelope is embedded with glycoprotein (G) spikes. A layer of a matrix protein (M) forms a bridge between G and the nucleocapsid proteins (N). Also included in the virion are several molecules of the RNA-dependent RNA polymerase (Large or L protein) and its cofactor, the phosphoprotein (P). The genes for these proteins are arranged as shown in the viral genome.

Acutely transforming retroviruses, which induce tumors in a short time period of weeks to months, carry modified versions of cellular oncogenes, called viral oncogenes. Slowly transforming retroviruses also subvert cellular oncogenes, but by integrating into or near the oncogene, thereby altering its expression, a process that can take years because of the apparently random nature of retrovirus integration.

Vaccines

Many viral infections can be prevented by vaccination. Several classes of vaccines are currently in use in humans and animals. Inactivated vaccines, such as the poliovirus vaccine developed by Jonas Salk, are produced from virulent viruses that are subjected to chemical treatments that result in loss of infectivity without complete loss of antigenicity (antigenicity is the ability to produce immunity). Another approach is the use of weakened variants of a virus with reduced **pathogenicity** to induce a protective immune response

pathogenicity ability to cause disease

without disease. While vaccines are usually given before exposure to a virus, postexposure vaccines can cure some virus infections with extended incubation periods, such as rabies.

Vaccines against smallpox eradicated the illness in 1980. It is believed that it may also be possible to eliminate polio. A recombinant vaccine against hepatitis B virus is now produced in yeast. However, developing effective vaccines to some viruses, including the common cold viruses, HIV-1, herpesviruses, and HPV, is proving very difficult principally due to the existence of many variants. Public health measures, such as mosquito control programs to curb the spread of viral diseases transmitted by these **vectors**, and safe-sex campaigns to slow the spread of sexually transmitted diseases, can also be effective. Because viruses replicate in cells, drugs that target viruses typically also affect cell functions. These therapeutic agents must be active against the virus while having “acceptable toxicity” to the host organism. The majority of the specific antiviral drugs currently in use target viral enzymes. For example, **nucleoside** analogues that target viral polymerases are active against HIV and certain herpesviruses. **SEE ALSO** BIOTECHNOLOGY; CANCER; CELL, EUKARYOTIC; DNA POLYMERASES; GENE THERAPY; HIV; ONCOGENES; RETROVIRUS; TRANSFORMATION; TUMOR SUPPRESSOR GENES; VIROIDS AND VIRUSOIDS.

vectors carriers

nucleoside building block of DNA or RNA, composed of a base and a sugar

Robert F. Garry

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Watson, James

Geneticist
1928–

James Dewey Watson was the codiscoverer of the structure of DNA. He has also made major contributions to research in genetics and molecular biology as an administrator, and has written widely read and influential books for both academic and nonscience audiences.

Early Life and Training

Watson was born April 6, 1928, in Chicago, Illinois. He showed his brilliance early, finishing high school in two years and appearing as one of the original “Quiz Kids,” on a popular 1940s radio show of the same name. He was graduated from the University of Chicago in 1947 with a B.S. in zoology, reflecting an early love of birds. He did his doctoral work at Indiana University in genetics, and earned a Ph.D. in 1950. He was drawn to Indiana by the chance



to work with Hermann Joseph Muller, who had been one of Thomas Hunt Morgan's associates in the famous "fly room" at Columbia University, and who had received a Nobel Prize for his discoveries in genetics. Watson's thesis adviser and principal mentor was Salvador Luria, who, along with Max Delbrück, had established bacterial genetics as the experimental system in which most of the major discoveries in molecular biology were to be made. Watson's thesis was on the effect of X rays on the multiplication of a bacterial virus, called phage.

Watson continued to study phage as a postdoctoral student in Copenhagen, Denmark where he worked from 1950 to 1951. While there, he met Maurice Wilkins, and for the first time saw the X-ray diffraction images generated in Wilkins's lab by Rosalind Franklin. Watson quickly decided to turn his attention to discovering the structure of important biological molecules, including DNA and proteins. By that time, DNA had been shown to be the genetic molecule, and it was believed that it somehow carried the instructions for making proteins, which actually perform most of the work in a cell.

The Structure of DNA

Luria arranged for Watson to continue his work at the Cavendish Laboratory in Cambridge, England, which was a center for the study of biomolecular structure, and Watson arrived there in late 1951. At the Cavendish, he met Francis Crick, who, after training in physics, had turned his attention to similar structural questions. The two hit it off, and began collaborating on the structure of DNA.

Watson and Crick approached the problem by building models of the four nucleotides known to make up DNA. Each was composed of a sugar called deoxyribose, a phosphate group, and one of four bases, called adenine, thymine, cytosine, and guanine. They knew the sugars and phosphates alternated to form a chain, with the bases projecting off to the side. The X-ray images they had seen suggested the structure was a helix, and offered more information about dimensions as well. They also knew that the biochemist Erwin Chargaff had discovered that the amounts of adenine and thymine in a cell's DNA were equal, as were the amounts of cytosine and guanine.

After several failed attempts, more analysis of the X-ray images, and a fortuitous conversation with a biochemist who corrected one of their hypothesized base structures, they developed the correct model. The helix is formed from two opposing strands of sugar phosphates, while the bases project into the center. Weak bonding (called **hydrogen bonding**) between bases holds them together. The key, as Watson and Crick discovered, was that the hydrogen bonds work best when adenine pairs with thymine, and guanine with cytosine, thus explaining Chargaff's ratios. The structure immediately suggested a replication mechanism, in which each side serves as the **template** for the formation of a new copy of the opposing side, and they speculated, correctly, that the sequence of the bases was a code for the sequence of amino acids in proteins. They published their results in 1953, and received the Nobel Prize for physiology and medicine for it 1962, along with Wilkins (Franklin by then had died, and was therefore ineligible for the prize).

hydrogen bonding weak bonding between the H of one molecule or group and a nitrogen or oxygen of another

template a master copy

Later Accomplishments

Watson remained active in the study of DNA and RNA for a number of years after the publication of the DNA structure. He joined the faculty of Harvard University in 1955, and remained there until 1976. During this time, he wrote an influential textbook, *Molecular Biology of the Gene*, and an enormously popular (and colorful) account of his and Crick's discovery, called *The Double Helix*.

In 1968 Watson became the director of the Cold Spring Harbor Laboratory on Long Island, New York, and he became president of the laboratory in 1994, a position he continues to hold. Watson revitalized this laboratory, helping it become one of the premier genetics research institutions in the world. His organizational drive was also called upon in 1988, when he spearheaded the launch of the U.S. Human Genome Project, dedicated to determining the sequence of the entire three billion bases in the genome. He headed the project from 1988 to 1992.

Throughout his career, Watson has invariably been described as “brash,” reflecting his capacity to take on big projects and big ideas, and his enthusiasm for making daring, occasionally outrageous predictions about the causes of an unexplained phenomenon or the direction science will take. Explaining this tendency in relation to his work on DNA, Watson wrote, “A potential key to the secret of life was impossible to push out of my mind. It was certainly better to imagine myself becoming famous than maturing into a stifled academic who had never risked a thought.” SEE ALSO CRICK, FRANCIS; DELBRÜCK, MAX; DNA; DNA STRUCTURE AND FUNCTION, HISTORY; MORGAN, THOMAS HUNT; MULLER, HERMANN; NUCLEOTIDE.

Richard Robinson

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X Chromosome

The X chromosome occupies an exceptional place in the mammalian **genome**. Together with the Y chromosome, the X chromosome differentiates the sexes. Males have one X chromosome and a Y chromosome and females have two X chromosomes. Because of this fundamental genetic difference, diseases caused by genes located on the X chromosome affect males and females differently and thus present unusual inheritance patterns. Furthermore, equal dosage of expression from genes on the X chromosome is restored between males and females by a special process called X inactivation, in which genes on one of the female X chromosomes are shut down.



genome the total genetic material in a cell or organism

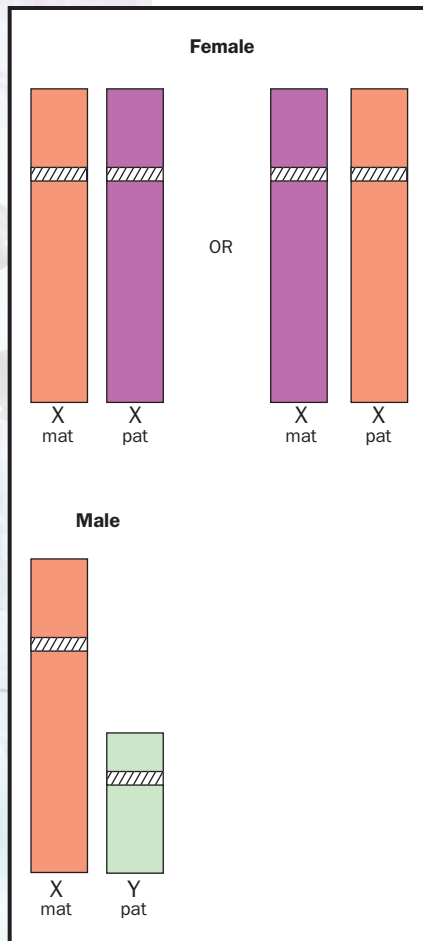


Figure 1. Schematic representation of the sex chromosomes of a female and male. The active X chromosome is in orange, the inactive X chromosome in purple, and the Y chromosome in green. In females, either the maternal (mat) or paternal (pat) X chromosome is inactivated in any given cell.

recombining exchanging genetic material

germ cells cells creating eggs or sperm

Role of the X Chromosome in Sex Differentiation

The so-called sex chromosomes differentiate the sexes: females are XX and males XY, which is the basis for the development of a fetus into a girl or a boy (Figure 1). All other chromosomes (called autosomes) are present in two copies in both males and females. It is the presence of a Y chromosome that determines the male sex of a baby, because the Y carries a gene that induces undifferentiated gonads to turn into testes in the fetus. The number of X chromosomes does not change the sex of a baby. Indeed, people with a single X chromosome and no Y chromosome are females with Turner syndrome, a rare genetic disorder characterized by short stature and infertility. Conversely, people who have two X chromosomes and a Y chromosome are males with Klinefelter's syndrome, which includes tall stature and infertility.

Sex Chromosome Evolution

Present-day sex chromosomes look very different from each other: The X chromosome comprises about 5 percent of the human genome, and contains about 2,000 genes, while the Y chromosome is quite small and contains only about 50 genes (Figure 1). This striking difference in size and gene content between the sex chromosomes makes it hard to believe that they are actually ancient partners in a pair of chromosomes that originally were very similar. Once sex became determined by a genetic signal from the Y, the sex chromosomes largely stopped **recombining** in **germ cells**. Degeneration of Y genes ensued, together with accumulation of genes that are advantageous to males on the Y chromosome, such as genes involved in testicular function and in male fertility. Similar genes appear to have accumulated on the X chromosome, so that the X chromosome also plays an important role in sperm production. The X chromosome may also have a prominent role in brain function and intelligence. A strong argument in favor of this intriguing but still controversial theory is that mental disability is more common in males.

The X Chromosome and Diseases

Some diseases affect males but not females in a family. Such diseases, called X-linked recessives, are often caused by mutations in genes located on the X chromosome, called X-linked genes. An X-linked disease is transmitted from the mother, not from the father, to an affected male, and an affected male will transmit a copy of the mutant gene to all his daughters. A famous example of an X-linked disease is hemophilia A. The blood of hemophiliac males fails to coagulate properly, leading to thinning of the blood and unstoppable bleeding after injury. This disease was recognized in the royal family of Queen Victoria, where examination of the huge pedigree readily confirmed recessive X-linked inheritance. Only males were affected, having inherited an X chromosome with a copy of a mutated gene from their healthy mothers, who were carriers of the disease. The mutated gene in hemophilia A was identified as factor VIII, a gene that encodes a protein essential for proper clotting of the blood. Males with a mutated gene cannot compensate since they have only one X chromosome, whereas female carriers have one normal gene that can compensate for the diseased gene. This typical recessive X-linked inheritance has been described for a variety of genes.

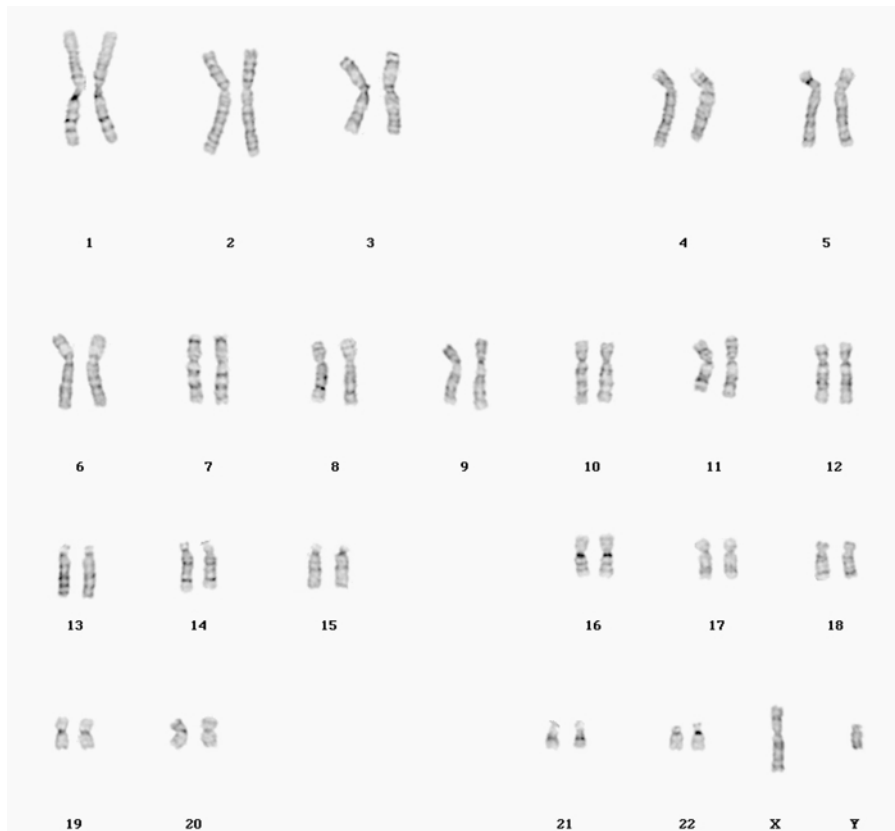


Figure 2. Karyotype of a normal human male. Each of the first twenty-two chromosomes has two copies. The last set is the sex chromosomes, and consists of one X and one Y chromosome.

Dominant X-linked mutations, in which female carriers with just one mutated copy of the gene are affected, are rare. One example of such a disease is vitamin D-resistant rickets, in which people develop skeletal deformities. Generally, the disease is less severe in females than in males, because of X inactivation (see below). A famous X-linked disorder with inheritance that cannot be classified as either recessive or dominant is fragile X mental retardation. The fragile X chromosome bears its name because it displays a site susceptible to chromosome breakage. The mutated gene at the site contains a triplet repeat expansion, in which a series of three consecutive bases are copied multiple times. This causes the gene to be turned off by secondary changes in its structure. Affected males have severe mental retardation and female carriers can also be affected.

X Inactivation

X inactivation consists of the silencing of genes on one of the X chromosomes in the female fetus. This silencing, which results in the absence of protein products from the inactivated genes, restores equal X-linked gene expression between the sexes. So, in the end, females have only one active X chromosome, like males (Figure 1). In the case of individuals with an abnormal number of X chromosomes, such as three X chromosomes, only one X will remain active.

One may wonder then why females do not express **deleterious** recessive X-linked mutations like males. This is because X inactivation is random, and a female is a mosaic of cells with either her paternal X active or her maternal X active (Figure 1). Thanks to this randomness, female carriers usually

deleterious harmful

epigenetic not involving
DNA sequence change



diploid possessing
pairs of chromosomes,
one member of each
pair derived from each
parent

haploid possessing only
one copy of each chro-
mosome

have plenty of cells with the normal gene remaining functional. Sometimes, there is even cell selection in carrier females, leading to skewed X inactivation in favor of the normal gene remaining functional. One intriguing feature of X inactivation is that it does not affect all X-linked genes. About 20 percent of genes “escape” X inactivation in humans. With a higher expression level in females than in males, such genes could perhaps play a role in female-specific functions. In males, some of these unusual genes have retained a functionally similar gene on the Y, as remnants of the ancient partnership of the sex chromosomes.

X inactivation was discovered in 1961 by Mary Lyon, a British scientist who studied mice. Thus, another name for this phenomenon is “Lyonization.” The physiologic or normal regulation of expression of many genes is at the level of the individual gene. In contrast, X inactivation regulates a whole chromosome that comprises a huge number of genes. Special mechanisms of regulation evolved to initiate X inactivation through the action of a master gene on the X. Once one of the two X chromosomes (maternal or paternal) is randomly chosen to become inactivated in a given fetal cell, it will be faithfully maintained in this state in the progeny of the cell. The stability of the inactivation is mediated by a series of complex molecular changes called **epigenetic** modifications. X inactivation is lost in only one type of cells, the female germ cells, where both X chromosomes are functional for transmission to the next generation. Thus, X inactivation involves special mechanisms of initiation, maintenance, and reactivation. Much work still needs to be done to fully understand the fascinating roles of the X chromosome and its regulation. **SEE ALSO** CHROMOSOMAL ABERRATIONS; FRAGILE X SYNDROME; HEMOPHILIA; INHERITANCE PATTERNS; INTELLIGENCE; MEIOSIS; MOSAICISM; SEX DETERMINATION; Y CHROMOSOME.

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Y Chromosome

The **diploid** human genome is packaged within 46 chromosomes, as two pairs of 23 discrete elements, into all cells other than the **haploid** gametic egg and sperm cells. During the reproductive process, each parent’s gametes contribute 22 nonsex chromosomes and either one X or one Y chromosome.

Paternal Inheritance

The X and Y chromosomes are the sex chromosomes for mammals, including humans. Not only are the X and Y sex chromosomes in mammals physically distinctive, with the Y being smaller, the Y chromosome is exceptionally peculiar. The X chromosome contains considerably more genes than the Y, which has its functionality essentially limited to traits associated with being

male. It is the Y chromosome that carries the major masculinity-determining gene (*SRY*, for sex-determining region Y), which dictates maleness. In a mating pair, if the paternal partner contributes a normal Y chromosome, male gonadal tissues (testes) develop in the offspring. Only males have the potential to transmit a Y chromosome to the next generation, and thus the father's contribution is decisive regarding an offspring's sex.

Since normally only one Y chromosome exists per cell, no pairing between X and Y occurs at meiosis, except at small regions. Normally, no crossing over occurs. Therefore, except for rare mutations that may occur during spermatogenesis, a son will inherit an identical copy of his father's Y chromosome, and this copy is also essentially identical to the Y chromosomes carried by all his paternal forefathers, across the generations. This is in contrast to the rest of his chromosomal heritage, which will be a unique mosaic of contributions from multiple ancestors created by the reshuffling process of recombination.

Sex Determination and Y Chromosome Genes

While *SRY* is the most dramatic gene affiliated with the Y chromosome, about thirty other genes have been identified. Some notable representatives include *AZF**a*, *b*, and *c*, which are associated with spermatogenesis and male infertility, *SMCY*, associated with the immune response function responsible for transplantation rejection when male tissue is grafted to female tissue, and *TSPY*, which may play a role in testicular cancer.

Sex Chromosome Evolution and Peculiarities

Discussions of sex chromosome evolution raise the question of the biological risks and benefits of sexual differentiation in organisms. Overall, sexual **dimorphism** enhances diversity that, in turn, improves the chances for evolutionary change and potential survival during periods of environmental change.

There are risks in the specialization of the Y chromosome, however. Besides its absence in females, lack of recombination for most of its physical territory except at its tips, and the strict pattern of paternal inheritance, the solitary cellular existence of the Y chromosome reduces the opportunity for DNA repair, which normally occurs while pairing during mitosis. This may explain the prevalence of multicopy DNA sequences on the Y, and why many of its genes have lost functionality. In fact, while genes predominately specific to male function tend to accumulate on the Y chromosome, other genes that have functional counterparts elsewhere will atrophy over evolutionary time, through the accumulation of uncorrected mutations. Thus the Y chromosome is slowing evolving toward a composition with fewer and fewer essential genes.

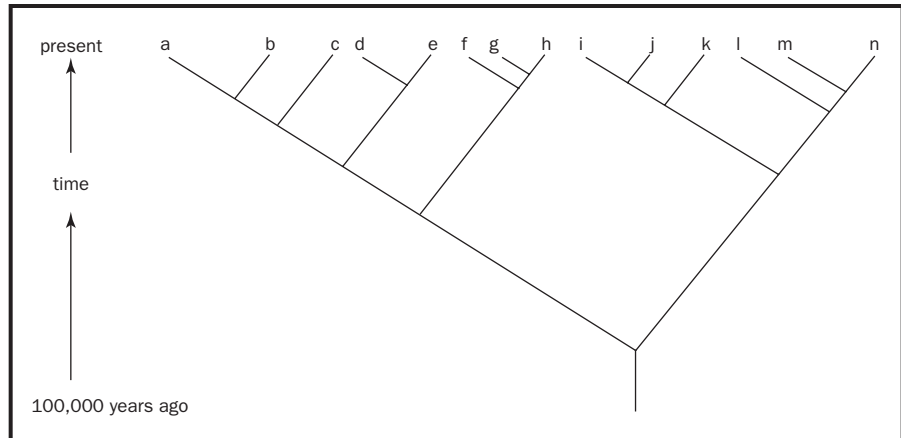
Molecular Anthropology Using the Y Chromosome

The field of molecular anthropology is predicated on the concept that the genes of modern populations encode aspects of human history. By studying the degree of genetic molecular variation in modern organisms, one can, in principle, understand past events. The Y chromosome is uniquely suited to such studies. Secondary applications of Y chromosome variation studies

dimorphism two forms



Schematic illustration of a gene tree created using Y chromosome polymorphisms. Each modern population (a–n) is descended from a single ancestral population living 100,000 years ago. Each branching point represents a mutation event, which is then faithfully inherited.



include forensics (criminological investigations, such as determining whether or not an individual has been involved in a crime) and genealogical reconstruction (verifying membership in a particular family's ancestry).

DNA polymers (such as chromosomes) are composed of a four-letter alphabet of chemicals called nucleotide bases. Random unique event mutations in DNA sequences can change the identity of a single base in the DNA molecule. These “spelling changes” are the essential currency of genetic anthropological research.

What is central is the assumption that a particular mutation arose just once in human history, and all men that display such a mutation on their Y chromosome descend from a common forefather on whom the mutation first appeared. The sequential buildup of such mutational events across the generations can be readily determined and displayed as a gene tree. Informally, the last known mutation to accumulate on a particular chromosome can be used to define a particular lineage or branch tip in the tree. As long as the mutational change does not affect the individual's ability to reproduce, it may be preserved and handed down to each succeeding generation, eventually becoming widespread in a population. Such mutations are called **polymorphisms** or genetic markers.

Since most of the Y chromosome has the special property of not recombining during meiosis, no shuffling of DNA from different ancestors occurs. As a consequence, any Y chromosome accumulates all the mutations that have occurred during its lineal life span and thus preserves the paternal genetic legacy that has been transmitted from father to son over the generations. The discovery of numerous Y chromosome polymorphisms has allowed us to deduce a reliable genealogy composed of numerous distinctive lineages. This concept is analogous to the genealogical relationships maintained by the traditional transmission of surnames in some cultures, although the gene tree approach provides access to a prehistorically deeper set of paternal relationships.

Molecular anthropologists have exploited this knowledge in an attempt to understand the history and evolutionary relationships of contemporary populations by performing a systematic survey of Y-chromosome DNA sequence variation. The unique nature of Y-chromosome diversification provides an elegant record of human population histories allowing

polymorphisms DNA sequence variant

researchers to reconstruct a global picture, emblematic of modern human origins, affinity, differentiation, and demographic history. The evidence shows that all modern extant human Y chromosomes trace their ancestry to Africa, and that descendants left Africa perhaps less than 100,000 years (or approximately 4,000 generations) ago.

While variation in any single DNA molecule can reflect only a small portion of human diversity, by merging other genetic information, such as data from the maternally transmitted mitochondrial DNA molecule, and nongenetic knowledge derived from archeological, linguistic, and other sources, we can improve our understanding of the affinities and histories of contemporary peoples. SEE ALSO MOLECULAR ANTHROPOLOGY; POLYMORPHISMS; SEX DETERMINATION; X CHROMOSOME.

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Yeast

Yeast are single-celled eukaryotic organisms related to fungi. The baker's yeast *Saccharomyces cerevisiae* and the distantly related *Schizosaccharomyces pombe* are favored model organisms for genetic research. The interest in yeast research stems from the fact that, as eukaryotic organisms, the sub-cellular organization of yeast is similar to that of cells of more complex organisms. Thus, understanding how a particular gene functions in yeast frequently correlates to how similar genes function in mammals, including humans.

Yeast Genetics

Yeast have many advantages as a genetic research tool. First, yeast are non-pathogenic (they do not cause diseases) and are therefore easy and safe to grow. Yeast can divide by simple fission (mitosis) or by budding and, like bacteria, they can be rapidly grown on solid **agar** plates or in liquid **media**. After just a few days in culture, a single yeast cell can produce millions of identical copies of itself, giving scientist a large supply of a genetically pure research tool.

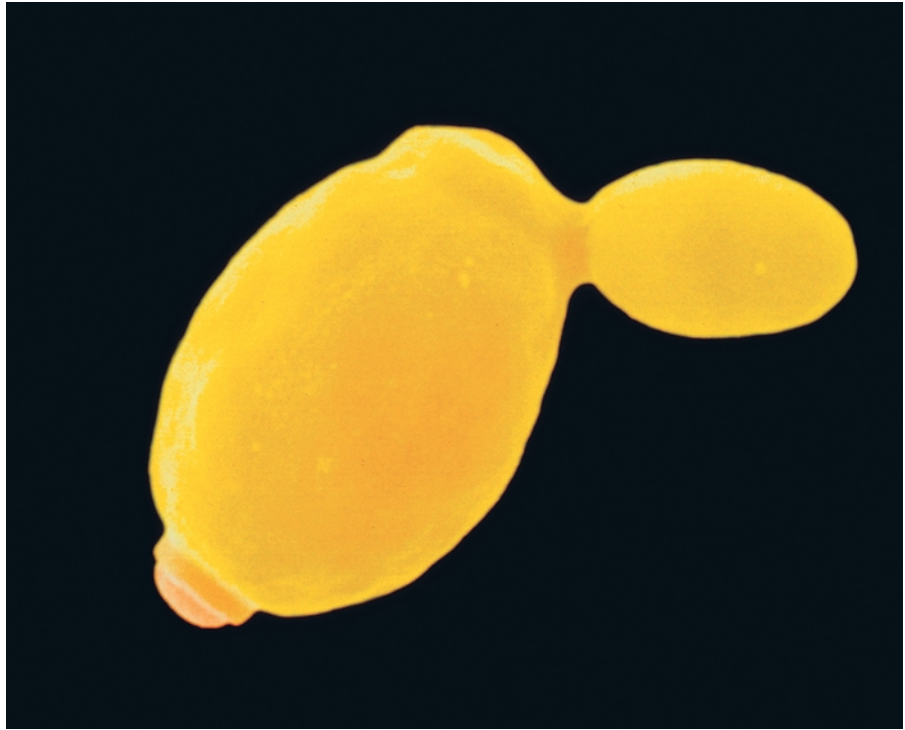
Second, yeast grow as either haploids (having only one set of chromosomes) or diploids (with two chromosome sets). Thus, genetically recessive mutations can be readily identified by **phenotypic** (visually observable) changes in the haploid strain. In addition, complementation can be performed by simply mating two haploid strains, where one does not contain the mutation. The resulting diploid strain contains both the functional and nonfunctional version of a gene responsible for a phenotype. The addition

agar gel derived from algae

media nutrient source

phenotypic related to the observable characteristics of an organism

As shown by a colorized scanning electron micrograph of a yeast cell magnified 3,025 times, this single-celled eukaryotic fungus multiplies by budding.



of the functional gene complements for the defect caused by the nonfunctional gene in the haploid strain. Diploid strains can be induced to undergo meiosis, a process in which the cell divides and passes one-half of its chromosomes to each of the resulting cells. After two such divisions, reproductive structures called asci are produced that contain four haploid offspring, called ascospores. The asci can be dissected and each of the ascospores isolated. In this way, scientists can easily mate different yeast strains and obtain new haploid genotypes through sexual reproduction and meiosis.

Third, the genome of yeast is small, about 3.5 times larger than that of bacteria and 200 times smaller than that of mammals. The yeast genome is arranged in 16 linear chromosomes that range from 200 to 2,200 kilobases in length. Unlike mammals, the yeast genome is very compact, with only 12 million base pairs, very few **introns**, and very little spacer DNA between functional genes. As a result, in 1996 baker's yeast was the first eukaryotic organism to have its entire genome sequenced.

introns untranslated portions of genes that interrupt coding regions

Genetic Transformation

Finally, one of the most useful properties of yeast for genetic studies is the ease with which DNA can be introduced into them, in a process called transformation. The introduced DNA can be maintained on self-replicating, circular strands of DNA called plasmids, or it can integrate into the yeast genome. Most importantly, integration usually occurs by a process called homologous recombination, whereby the introduced DNA replaces chromosomal DNA that contains the same sequence. This process permits scientists to readily mutate any yeast gene and replace the native gene in the cells with the mutated version. Since yeast can be grown as haploids, the phenotypic changes caused by the introduced gene can be readily identified.

In addition, the function of a cloned piece of DNA (e.g., a gene) can be identified by transforming yeast in which the DNA is carried on a circular plasmid. The introduced gene may either functionally replace a defective gene or cause a phenotypic defect in the cells indicating a function for that gene.

The ability to complement yeast defects with cloned pieces of DNA has been extended to mammalian genes. Recognizing that some genes have similar sequences and functions in both mammals and yeast, scientists sometimes use yeast as a tool to identify the functions of mammalian genes. Not many mammalian genes can directly substitute for a yeast gene, however. More frequently, scientists study the yeast gene itself to understand how its protein functions in the cell. The knowledge gained can often lead to an understanding of how similar genes might function in mammals. Now that the yeast genome has been completely sequenced and the results have been deposited in a public databank for all to use, rapid progress is being made in identifying all yeast genes and their functions.

An important method for studying mammalian genes in yeast is called the two-hybrid system. This system is used to determine if two proteins functionally interact with each other. Both genes are cloned into yeast plasmids and transformed into the cells. A special detection system is used that is active only when both cloned proteins physically contact each other in the cell. When that happens, scientist can identify which proteins need to interact with each other in order to function.

Yeast are also being used in the laboratory and commercial production of important nonyeast proteins. Foreign genes are transformed into yeast and, after **transcription** and translation, the foreign proteins can be isolated. Because of the ease of growing large quantities of cells, yeast can produce a large amount of the protein. While similar protein production can be performed by bacteria, eukaryotic proteins often do not function when made in bacteria. This is because most eukaryotic proteins are normally altered after translation by the addition of short sugar chains, and these modifications are often required for proper function, but bacteria do not carry out these necessary post-translational modifications. Yeast, however, does permit these modifications, and is thus more likely to produce a functional protein. SEE ALSO CELL, EUKARYOTIC; CELL CYCLE; GENOME; HUMAN GENOME PROJECT; MODEL ORGANISMS; PLASMID; POST-TRANSLATIONAL CONTROL; TRANSFORMATION; TRANSGENIC ANIMALS.

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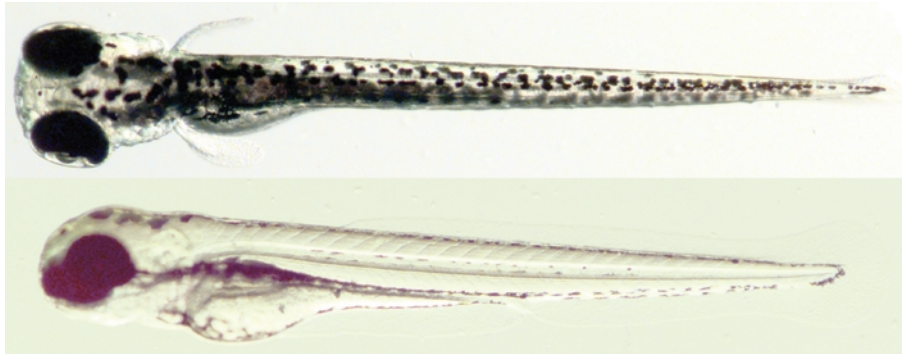
transcription messenger RNA formation from a DNA sequence

Zebrafish

The zebrafish (*Brachydanio rerio*) is a small tropical freshwater fish that began to be used as a genetic model system in the early 1980s. The zebrafish shares numerous anatomical and genetic similarities with higher vertebrates,



The developing zebrafish is transparent, making it an ideal subject for the study of development.



embryogenesis development of the embryo from a fertilized egg

histological related to tissues

including humans, both in the general body plan and in specific organs. Close parallels exist in many aspects of early **embryogenesis** and in the anatomical and **histological** features of the brain, spinal cord, sensory systems, cardiovascular system, and other organs. Not infrequently, genetic defects in zebrafish resemble human disorders. Owing to these similarities, zebrafish genetics is being broadly applied to address both basic biological questions and to model human inherited diseases.

Useful Traits for Researchers

Several characteristics favor the choice of zebrafish for genetic research. First, zebrafish are easy to maintain in large numbers in a small laboratory space. Second, their generation time is relatively short: three months. Finally, females produce large clutches of offspring, about 50 to 100 a week. The zebrafish also presents advantages for embryological analysis: Its embryos develop externally, and are largely transparent for the first 36 hours of development.

The first proponents of the zebrafish model, George Streisinger and his colleagues at the University of Oregon, outlined its genetic characteristics and provided a thorough description of zebrafish embryogenesis. Zebrafish research entered a new phase when two groups, at the University of Tuebingen (Germany), and at Harvard Medical School, performed large-scale chemical **mutagenesis** experiments and isolated nearly 2,000 mutations that affect almost every aspect of embryonic development, from **gastrulation** to axonal pathfinding. Once the mutations were identified, they were studied to determine their developmental effects, after which their genetic and molecular nature was analyzed. These early genetic studies in zebrafish were limited by the lack of genomic resources, such as maps of the genome or genomic libraries, but these deficiencies were gradually eliminated during the late 1990s. Zebrafish research has entered a new phase since the completion of the genome project. This effort provided the sequence of the entire zebrafish genome and made cloning of zebrafish genes much more efficient.

Ongoing Research Involving Zebrafish

Zebrafish research continues at a fast pace. New rounds of mutation identification are in progress. To supplement chemical mutagenesis, retroviral vectors were developed as mutagenic agents and applied on a large scale. Tools to study the functions of individual genes, such as **transgenics**, were

mutagenesis creation of mutations

gastrulation embryonic stage at which primitive gut is formed

transgenics transfer of genes from one organism into another

developed in parallel to mutant screening approaches. A technique that complements mutagenesis screens in zebrafish very well is known as gene knock-down. This approach uses modified antisense oligonucleotides to block the function of specific genes. These oligonucleotides contain a substitution of the sugar ring in the nucleic acid backbone that makes them resistant to degradation by enzymes in living tissues. Gene knockdown is widely used to study mutant **phenotypes** of genes for which chemically induced mutations are not available.

phenotypes observable characteristics of an organism

The usefulness of the zebrafish as a model organism originates in its unique combination of genetic and embryological characteristics. Genetic approaches, such as mutagenesis screens, can be combined in zebrafish with other techniques, enabling researchers to study cell movements, cell birth dates, or interactions between cells in the living embryo. Although most of the zebrafish genetic research focuses on embryonic development, other problems, such as the genetic basis of circadian rhythms, cancer formation, neurodegenerative disorders, and drug addiction, are also being addressed. SEE ALSO DNA LIBRARIES; MODEL ORGANISMS; MUTAGENESIS; TRANSGENIC ORGANISMS; ETHICAL ISSUES.

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Glossary

α the Greek letter alpha

β the Greek letter beta

γ the Greek letter gamma

λ the Greek letter lambda

σ the Greek letter sigma

E. coli the bacterium *Escherichia coli*

“-ase” suffix indicating an enzyme

acidic having the properties of an acid; the opposite of basic

acrosomal cap tip of sperm cell that contains digestive enzymes for penetrating the egg

adenoma a tumor (cell mass) of gland cells

aerobic with oxygen, or requiring it

agar gel derived from algae

agglutinate clump together

aggregate stick together

algorithm procedure or set of steps

allele a particular form of a gene

allelic variation presence of different gene forms (alleles) in a population

allergen substance that triggers an allergic reaction

allolactose “other lactose”; a modified form of lactose

amino acid a building block of protein

amino termini the ends of a protein chain with a free NH_2 group

amniocentesis removal of fluid from the amniotic sac surrounding a fetus, for diagnosis

amplify produce many copies of, multiply

anabolic steroids hormones used to build muscle mass



- anaerobic** without oxygen or not requiring oxygen
- androgen** testosterone or other masculinizing hormone
- anemia** lack of oxygen-carrying capacity in the blood
- aneuploidy** abnormal chromosome numbers
- angiogenesis** growth of new blood vessels
- anion** negatively charged ion
- anneal** join together
- anode** positive pole
- anterior** front
- antibody** immune-system protein that binds to foreign molecules
- antidiuretic** a substance that prevents water loss
- antigen** a foreign substance that provokes an immune response
- antigenicity** ability to provoke an immune response
- apoptosis** programmed cell death
- Archaea** one of three domains of life, a type of cell without a nucleus
- archaeans** members of one of three domains of life, have types of cells without a nucleus
- aspirated** removed with a needle and syringe
- aspiration** inhalation of fluid or solids into the lungs
- association analysis** estimation of the relationship between alleles or genotypes and disease
- asymptomatic** without symptoms
- ATP** adenosine triphosphate, a high-energy compound used to power cell processes
- ATPase** an enzyme that breaks down ATP, releasing energy
- attenuation** weaken or dilute
- atypical** irregular
- autoimmune** reaction of the immune system to the body's own tissues
- autoimmunity** immune reaction to the body's own tissues
- autosomal** describes a chromosome other than the X and Y sex-determining chromosomes
- autosome** a chromosome that is not sex-determining (not X or Y)
- axon** the long extension of a nerve cell down which information flows
- bacteriophage** virus that infects bacteria
- basal** lowest level



- base pair** two nucleotides (either DNA or RNA) linked by weak bonds
- basic** having the properties of a base; opposite of acidic
- benign** type of tumor that does not invade surrounding tissue
- binding protein** protein that binds to another molecule, usually either DNA or protein
- biodiversity** degree of variety of life
- bioinformatics** use of information technology to analyze biological data
- biolistic** firing a microscopic pellet into a biological sample (from biological/ballistic)
- biopolymers** biological molecules formed from similar smaller molecules, such as DNA or protein
- biopsy** removal of tissue sample for diagnosis
- biotechnology** production of useful products
- bipolar disorder** psychiatric disease characterized by alternating mania and depression
- blastocyst** early stage of embryonic development
- brackish** a mix of salt water and fresh water
- breeding analysis** analysis of the offspring ratios in breeding experiments
- buffers** substances that counteract rapid or wide pH changes in a solution
- Cajal** Ramon y Cajal, Spanish neuroanatomist
- carcinogens** substances that cause cancer
- carrier** a person with one copy of a gene for a recessive trait, who therefore does not express the trait
- catalyst** substance that speeds a reaction without being consumed (e.g., enzyme)
- catalytic** describes a substance that speeds a reaction without being consumed
- catalyze** aid in the reaction of
- cathode** negative pole
- cDNA** complementary DNA
- cell cycle** sequence of growth, replication and division that produces new cells
- centenarian** person who lives to age 100
- centromere** the region of the chromosome linking chromatids
- cerebrovascular** related to the blood vessels in the brain
- cerebrovascular disease** stroke, aneurysm, or other circulatory disorder affecting the brain

- charge density** ratio of net charge on the protein to its molecular mass
- chemotaxis** movement of a cell stimulated by a chemical attractant or repellent
- chemotherapeutic** use of chemicals to kill cancer cells
- chloroplast** the photosynthetic organelle of plants and algae
- chondrocyte** a cell that forms cartilage
- chromatid** a replicated chromosome before separation from its copy
- chromatin** complex of DNA, histones, and other proteins, making up chromosomes
- ciliated protozoa** single-celled organism possessing cilia, short hair-like extensions of the cell membrane
- circadian** relating to day or day length
- cleavage** hydrolysis
- cleave** split
- clinical trials** tests performed on human subjects
- codon** a sequence of three mRNA nucleotides coding for one amino acid
- Cold War** prolonged U.S.-Soviet rivalry following World War II
- colectomy** colon removal
- colon crypts** part of the large intestine
- complementary** matching opposite, like hand and glove
- conformation** three-dimensional shape
- congenital** from birth
- conjugation** a type of DNA exchange between bacteria
- cryo-electron microscope** electron microscope that integrates multiple images to form a three-dimensional model of the sample
- cryopreservation** use of very cold temperatures to preserve a sample
- cultivars** plant varieties resulting from selective breeding
- cytochemist** chemist specializing in cellular chemistry
- cytochemistry** cellular chemistry
- cytogenetics** study of chromosome structure and behavior
- cytologist** a scientist who studies cells
- cytokine** immune system signaling molecule
- cytokinesis** division of the cell's cytoplasm
- cytology** the study of cells
- cytoplasm** the material in a cell, excluding the nucleus



- cytosol** fluid portion of a cell, not including the organelles
- de novo** entirely new
- deleterious** harmful
- dementia** neurological illness characterized by impaired thought or awareness
- demography** aspects of population structure, including size, age distribution, growth, and other factors
- denature** destroy the structure of
- deoxynucleotide** building block of DNA
- dimerize** linkage of two subunits
- dimorphism** two forms
- diploid** possessing pairs of chromosomes, one member of each pair derived from each parent
- disaccharide** two sugar molecules linked together
- dizygotic** fraternal or nonidentical
- DNA** deoxyribonucleic acid
- domains** regions
- dominant** controlling the phenotype when one allele is present
- dopamine** brain signaling chemical
- dosage compensation** equalizing of expression level of X-chromosome genes between males and females, by silencing one X chromosome in females or amplifying expression in males
- ecosystem** an ecological community and its environment
- ectopic expression** expression of a gene in the wrong cells or tissues
- electrical gradient** chemiosmotic gradient
- electrophoresis** technique for separation of molecules based on size and charge
- eluting** exiting
- embryogenesis** development of the embryo from a fertilized egg
- endangered** in danger of extinction throughout all or a significant portion of a species' range
- endogenous** derived from inside the organism
- endometriosis** disorder of the endometrium, the lining of the uterus
- endometrium** uterine lining
- endonuclease** enzyme that cuts DNA or RNA within the chain
- endoplasmic reticulum** network of membranes within the cell

